

Protein-Protein Interaction Network Analysis for Endometriosis

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Abstract: We build the protein-protein interaction (PPI) network of endometriosis from 166 DisGeNET disease genes (UMLS concept C0014175), expanded through STRING first-shell neighbours to 454 nodes and 21,095 edges. Spread reduction is measured under SIR and Independent Cascade diffusion with calibrated parameters, and four blocking chains are built: two structural (greedy group degree and greedy iterative betweenness) and two diffusion-aware (greedy influence maximization under each model). Greedy iterative betweenness dominates, saturating near $k = 200$ at 99% spread reduction, while the diffusion-aware chains significantly underperform. The recommended protein set strongly depends on the algorithm: 93% of the top-20 picks are algorithm-specific. The PPI is 7.8 percentage points more resistant to betweenness blocking than its degree-randomized configuration null model, which is an evidence of structural redundancy more complex than the degree distribution. The diffusion-aware chains also surface the biology that the structural ones miss: angiogenesis and tissue-remodeling proteins under SIR, and immune-modulation proteins under IC, which are all relevant to endometriosis pathology.

Keywords: protein-protein interaction, endometriosis, group centrality, diffusion models, network blocking

1. INTRODUCTION

Endometriosis affects roughly 10% of women of reproductive age, but remains under-studied and under-funded: current treatment only targets symptoms instead of molecular drivers, and diagnosis is often delayed by years. Its prevalence and poor understanding make it a worthwhile target for a systems-level analysis.

Protein-protein interaction (PPI) networks represent the connections between proteins, which are the nodes, whereas edges represent physical or functional interactions, and the graph captures the structure through which biological signals propagate. On the disease subgraph, network science offers an interesting range of tools for reasoning about therapeutic targets: centrality identifies hubs and bridges, group centrality identifies sets whose joint removal disrupts connectivity, and diffusion models simulate how far a disease signal spreads and how much each intervention reduces it.

Using those tools we will try to answer the following research questions: which set of k proteins, blocked simultaneously, most limits propagation of the disease signal in the endometriosis PPI network, and does the optimal set depend on the chosen algorithm?

2. DATA AND NETWORK CONSTRUCTION

The gene list comes from DisGeNET², a curated repository of gene-disease associations, queried on UMLS concept C0014175 (endometriosis), human taxon 9606, curated provenance, and score at least 0.25. The query returns 166 genes, the seed pool for the network.

The interaction graph is built from STRING³ with first-shell expansion: STRING returns all interactions among the 166 genes plus their immediate neighbours, capped at `add_nodes = 300`, keeping only interactions with confidence at least 0.4, which is the medium-confidence default. The result is a single connected component of 454 nodes and 21,095 edges, with undirected, unweighted edges. First-shell expansion makes the graph unusually dense, with average degree near 93 and density about 0.205.

3. NULL MODELS

To test whether the structure and dynamics below reflect the PPI's topology rather than properties of any random graph of the same size, we compare it against three null models, each matched on $|V| = 454$.

The Erdős-Rényi model places each edge independently at the empirical density, fixing only node count and density, which makes it the weakest null model. The Barabási-Albert model is scale-free by preferential attachment, with $m \approx 53$ chosen for the edge count to match, isolating the effect of a scale-free degree distribution. The configuration model preserves the empirical degree sequence exactly and randomizes everything else, which makes it the strictest.

4. NETWORK STATISTICS

Table 1 compares the four networks on five standard descriptors; average degree is omitted, as it follows from edge and node counts.

Table 1. Network statistics for the empirical PPI and the three null models.

	$ E $	Density	Clustering	$\bar{\ell}$
PPI	21,095	0.205	0.637	1.94
ER	21,253	0.207	0.206	1.79
BA	21,253	0.207	0.298	1.79
CM	21,095	0.205	0.512	1.84

$\bar{\ell}$ denotes average shortest path length. ER and BA edge counts differ slightly from PPI because neither model locks the edge count exactly.

PPI differs from every null on two descriptors. Clustering is highest, at 0.637, against 0.206 (ER), 0.298 (BA), and 0.512 (CM); the configuration null comes closest, since preserving degrees induces some clustering, but PPI still exceeds it by about 25%, so its proteins form dense groups. Average shortest path length is similar in all networks, but slightly longer in PPI than the other models.

For the degree distributions (Figure 1), PPI is heavy-tailed, with many low-degree proteins and a few hubs spanning two orders of magnitude, and CM of course matches it by construction. BA has a power law distribution but, with minimum degree $m \approx 53$, misses the low-degree side, and ER is a narrow Poisson peak.

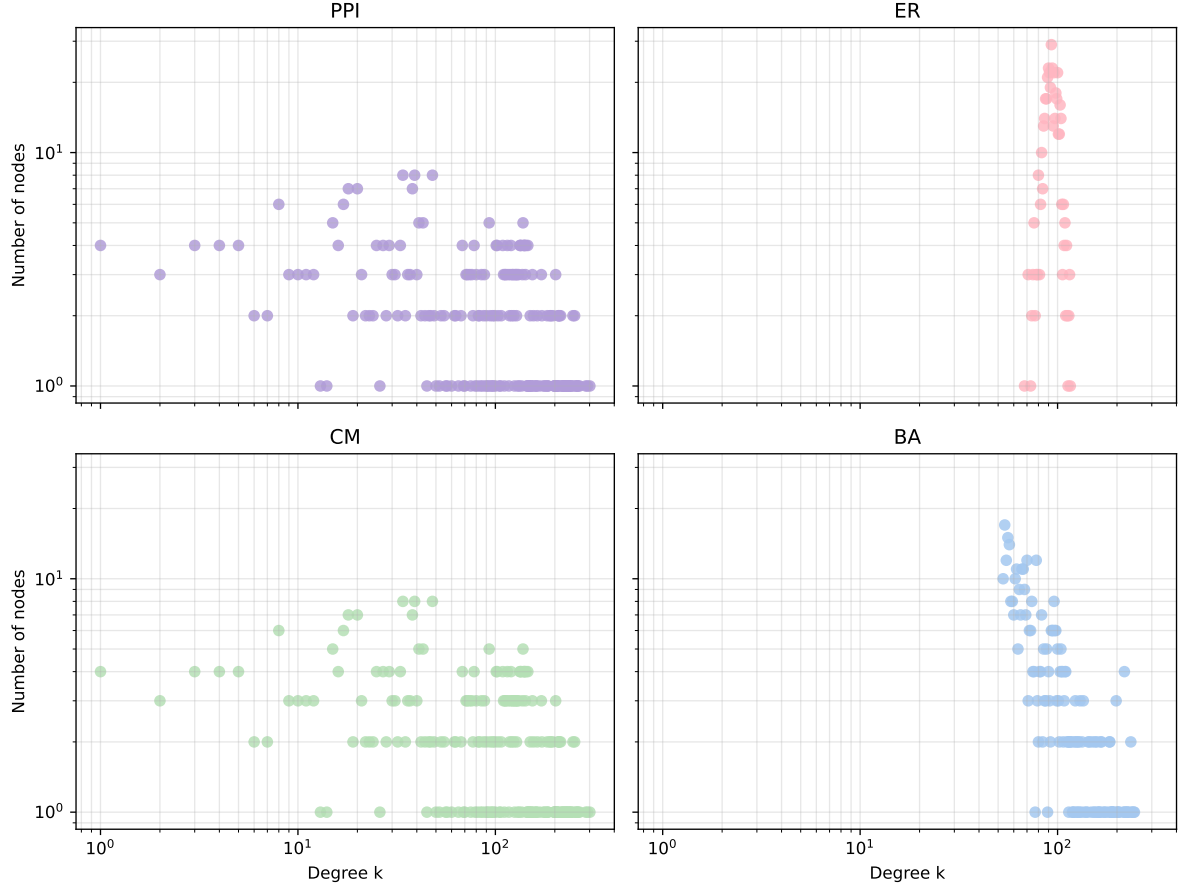
5. METHODOLOGY

We treat the diffusion models as measurement instruments and the blocking algorithms as the strategies that we test. For a candidate set B of size k , we simulate propagation with the nodes in B removed, average over Monte Carlo runs, and report the spread reduction relative to the unblocked baseline. The task is to choose the B of fixed size k that maximizes this reduction.

Two propagation models cover qualitatively different dynamics: SIR, the Susceptible-Infected-Recovered model, with infection rate β and recovery rate γ , and the Independent Cascade model, with activation probability p . Pairing single-shot IC with multi-attempt SIR tests whether the optimal set is sensitive to the dynamics.

The seeds are the 23 highest-confidence disease genes (those with DisGeNET score at least 0.7), which makes up about 5%

Figure 1: Degree distribution (log-log)

**Figure 1.** Degree distribution on log-log axes for the empirical PPI network and the three null models matched on node count.

of the network. Each configuration is averaged over $N_{\text{runs}} = 50$ runs. We calibrate β , γ , and p on the unblocked graph (Table 2). We use $\beta = 0.005$, $\gamma = 0.20$ for SIR and $p = 0.02$ for IC, giving unblocked spreads near 70% and 65% of the network.

Table 2. Calibration sweep on the unblocked PPI graph.

Parameter setting	Mean spread	% of network
SIR $\beta = 0.005, \gamma = 0.20$	318.7	70
SIR $\beta = 0.010, \gamma = 0.20$	382.3	84
SIR $\beta = 0.020, \gamma = 0.20$	418.6	92
SIR $\beta = 0.050, \gamma = 0.20$	439.5	97
IC $p = 0.005$	59.9	13
IC $p = 0.010$	170.5	38
IC $p = 0.020$	294.7	65
IC $p = 0.050$	386.0	85

Means over 50 Monte Carlo runs with the high-confidence seed set.

Three algorithms select the blocking set. Greedy group degree (gdeg) adds at each step the node whose closed neighbourhood covers the most uncovered nodes. Greedy iterative betweenness (gbtw) repeatedly removes the current highest-betweenness node, targeting the proteins whose removal most fragments the graph. Greedy influence maximization (gim) adds the node whose blocking most reduces simulated spread, which also makes it the only diffusion-aware strategy, run under both SIR and IC to give gim_sir and gim_ic.

6. RESULTS

6.1. Spread Reduction Across Strategies

Figure 2 plots spread reduction against blocked-set size for the four chains and a random baseline. gbtw definitely dominates: at $k = 200$ it reaches 98.7% under SIR and 99.1% under IC, with the next-best chain 12 to 17 points behind. It reaches saturation near $k = 200$, which is 44% of the network. After that, every chain keeps climbing mainly because the graph is shrinking, and even the random baseline reaches 86 to 89% by $k = 300$. The meaningful comparison is therefore $k \leq 200$.

Table 3 lists each chain’s top ten. The two structural chains share four of ten proteins, when every other pair shares none. Random size-ten subsets of a 454-node graph would overlap near zero, so the structural agreement, roughly 20 times that baseline, reflects a potential real signal, while the disjointness elsewhere does not.

Table 3. Top ten proteins selected by each strategy on the PPI graph.

	gdeg	gbtw	gim_sir	gim_ic
1	IL6	ALB	FGF4	CD40
2	ESR1	ESR1	KDR	LAG3
3	AKR1C3	EGFR	UGT1A7	CX3CL1
4	NCOA1	IL6	VTN	HDAC2
5	EGFR	GAPDH	IL17A	MED4
6	INS	TNF	NTRK3	UGT2B11
7	ITGB3	IL1B	PDGFRB	MED14
8	APOE	INS	SERPINF1	CCL13
9	CTNBN1	AKT1	EREG	PTPRC
10	CYP3A4	ACTB	TIMP1	COPS2

The diffusion-aware chains are significantly gbtw: gim_sir

Figure 2: Spread reduction vs blocked-set size, shaded band shows SEM

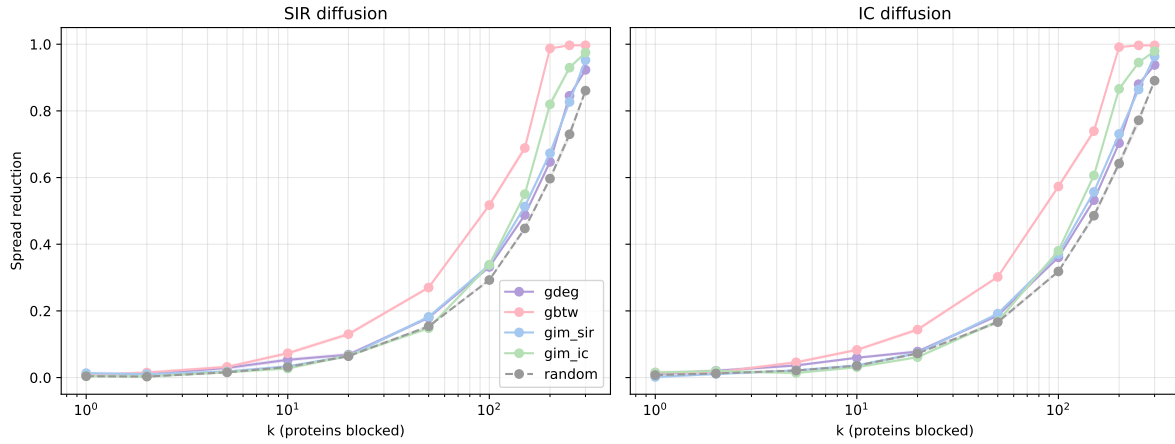


Figure 2. Spread reduction as a function of blocked-set size k for four selection chains and a random baseline. Left panel: SIR diffusion. Right panel: Independent Cascade. Shaded bands indicate one standard error of the mean.

Figure 3: Spread reduction on PPI vs null networks, shaded band shows SEM

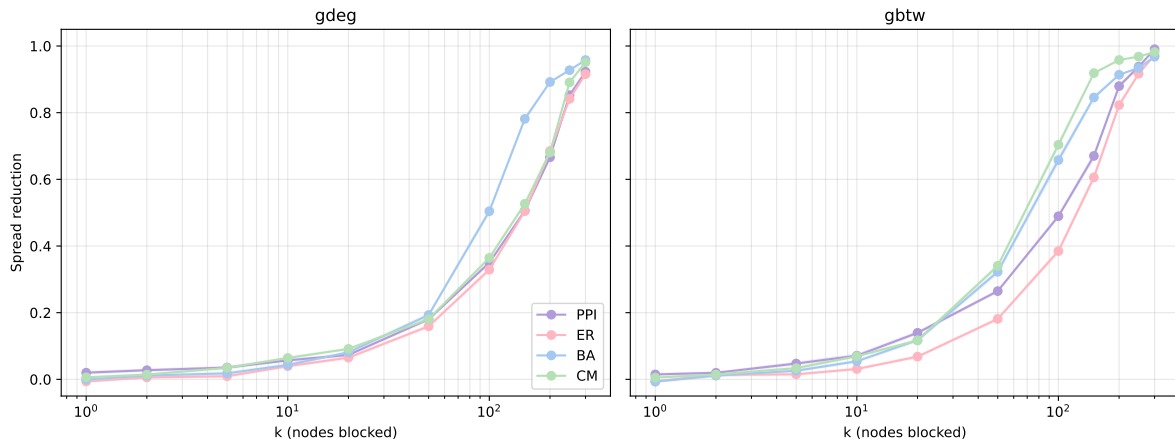


Figure 3. Spread reduction on PPI versus the three null networks under SIR diffusion. Left panel: greedy group degree. Right panel: greedy iterative betweenness. Shaded bands indicate one standard error of the mean.

reaches 33.7% at $k = 100$ against *gbtw*'s 51.7%, and the gap only widens with k raising. Although *gim* optimizes spread directly, two effects favour betweenness here. The dense graph puts high-betweenness hubs on many propagation paths at once, so betweenness is a strong proxy for disrupting diffusion, and *gim*'s marginal gains is probably caused by noisy Monte Carlo estimates that flatten the differences between very similar candidates.

6.2. PPI vs Null Networks

Figure 3 repeats the analysis on the null networks, this time using only *gdeg* and *gbtw*. The most interesting observation is PPI versus CM under *gbtw*: CM reaches 95.8% reduction at $k = 200$, PPI only 88.0%. Since the two share the exact same degree sequence, the 7.8 percentage-point gap must come from a more complicated structure that CM omits, like the clustering, which defends the PPI against targeted blocking.

Under *gdeg* the picture changes: PPI, ER, and CM are all near 67 to 68% and indistinguishable, while BA jumps to 89.2%. BA's few extreme hubs let degree-based blocking fragment it fast.

6.3. Top-k Composition

Figure 4 shows the union of the four top-20 lists as a rank heatmap. The union contains 75 proteins, where only 5 appear

in two chains and none in three or more, so 93% of the picks are algorithm-specific, which is the strongest piece of evidence for algorithm dependence. The only shared cluster is structural: *gdeg* and *gbtw* share ESR1, IL6, EGFR, INS, and CYP3A4. ESR1 is the estrogen receptor, crucial to an estrogen-dependent disease, while the others are general signaling hubs.

The diffusion-aware chains pick disjoint but biologically coherent sets. *gim_sir* leads with growth-factor and angiogenesis proteins (FGF4, KDR, PDGFRB, EREG) and extracellular-matrix targets (VTN, SERPINF1, TIMP1); *gim_ic* leads with immune-modulation proteins (CD40, LAG3, CX3CL1, CCL13, PTPRC) and transcriptional regulators (HDAC2, MED4, MED14). Both angiogenesis with ECM remodeling and immune dysregulation are documented axes of endometriosis pathology¹, and these proteins are absent from the structural chains, which might be because their database degree is lower than the canonical signaling hubs. The choice of algorithm therefore controls both the recommended proteins and the biology they point to.

7. DISCUSSION

Substantial spread reduction in this PPI requires removing roughly 200 of 545 proteins, far beyond what any realistic pharmacological intervention can target. At $k = 10$, the size of a generous multi-target therapy, even *gbtw* achieves only a minor

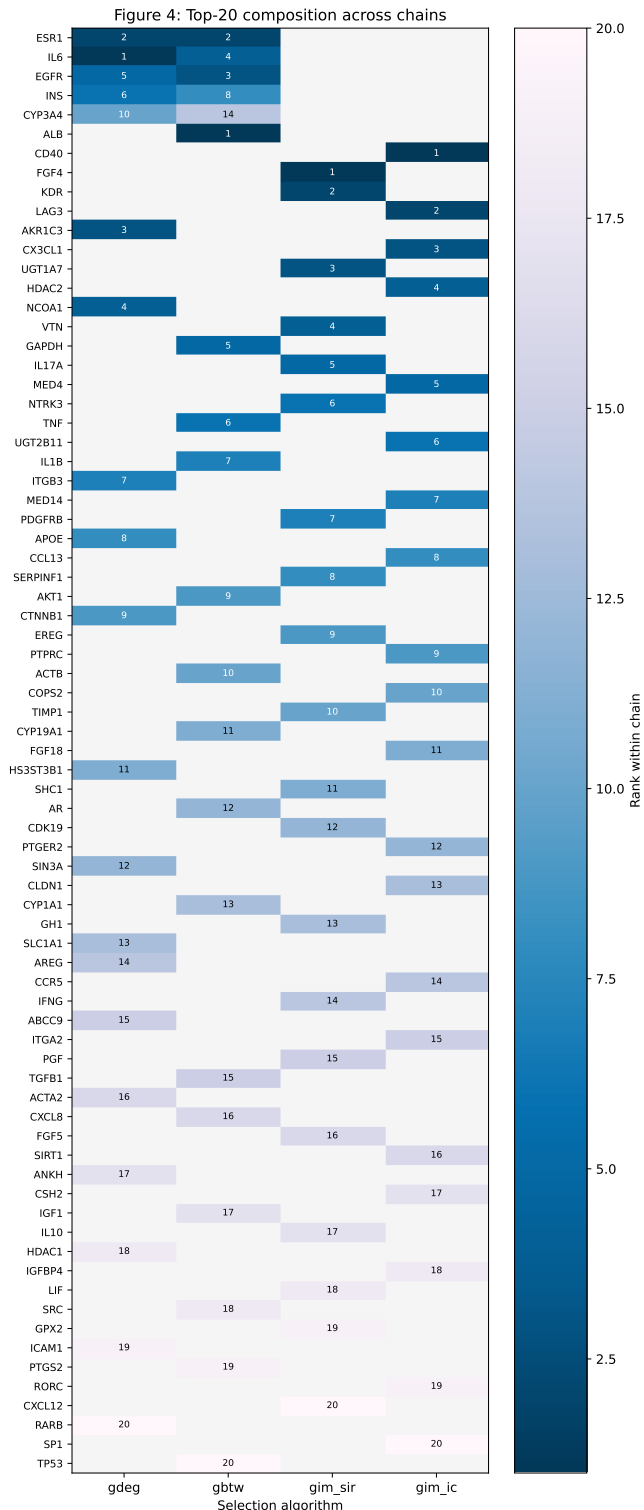


Figure 4. Top-20 union across the four selection chains. Each cell shows the within-chain rank when the protein is in that chain's top 20; blank means the protein is absent from that chain at this depth.

7% reduction under SIR. The PPI's clustering and modular structure buffer signaling against the removal of a small number of proteins, and the null comparison ties this buffering to the PPI's specific topology, since the random models do not show it.

The recommended set depends on the algorithm as much as on the biology, so the candidate protein lists are a more useful product than the spread number itself: the diffusion-aware chains surface angiogenesis and immune-modulation targets that align with known endometriosis pathology¹ and that topology alone misses entirely.

8. LIMITATIONS

A few caveats apply. `gim` used 25 inner Monte Carlo replicates rather than the 50 used for evaluation and was not re-run on the nulls, so the null comparison is one-sided; the STRING threshold of 0.4 also creates a dense graph that favours structural attacks.. More fundamentally, SIR and IC are structural proxies, because neither models biological signalling directly, so spread reduction is not a biological prediction and blocking a node is not the same as inhibiting a protein.

9. CONCLUSION

The optimal blocking set is the prefix of the greedy iterative betweenness chain, which saturates near $k = 200$, representing 44% of the graph. The optimal set depends strongly on the chosen algorithm: about 93% of the top-20 picks are algorithm-specific, and every pairwise top-ten comparison except `gdeg` versus `gbtw` shows zero overlap. At realistic small k the reduction is barely significant, so its main value lies in the candidate proteins it points to.

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